

Izge Bayyurt

25B Gilman Falls Ave, Old Town, ME 04468 - izge.bayyurt@maine.edu

207-216-5600 - izgebayyurt.github.io

Education

The University of Maine

Thesis title: Surface Explorer - Investigating large-scale 3D shapes using VR

Advisor: Justin Dimmel

M.Sc. in Teaching, 2025

GPA: 3.82

Colby College

Honors: *Davis UWC Scholar, Dean's List*

B.A. in Computer Science, 2022

GPA: 3.56

Research Experience

Immersive Mathematics in Rendered Environments Lab

Graduate & Research Assistant

The University of Maine

- Managed the lab under Justin Dimmel.
- Designed and developed 3 virtual reality (VR) environments in Unity and A-Frame to improve mathematical reasoning, 3D geometry, and spatial understanding in students.
- Conducted focus groups and individual interviews with undergraduate students and in-service teachers.
- Collected, transcribed and analyzed interviews and video recordings.
- Collaborated with other graduate and undergraduate students to design and program virtual environments.

INSITE Lab

Lead programmer & Designer

Colby College

- Worked with Stacy A. Doore to establish the lab.
- Designed and developed a 9x9x9 playable chessboard in VR and presented a poster in Colby Undergraduate Summer Research Retreat.
- Led the development a state-of-the-art offshore wind turbine farm simulation in VR in collaboration with National Renewable Energy Laboratory.
- Added physically simulated mooring lines and underwater representation to the offshore wind turbine simulation in the second rendition of the project.
- Co-authoring a paper with Stacy A. Doore and Dr. Alison W. Bates on our work with the wind turbine simulation and its use in engaging stakeholders in the Gulf of Maine.
- Mentored 7 undergraduate students by teaching them software development for VR and 3D design software and guided their work by holding regular meetings.

Conferences, Talks, & Posters

Talks

Justin K. Dimmel and Izge Bayyurt. "Rise of the Machines: Navigating the Opportunities and Challenges of AI-Assisted Research and Learning". In: *Proceedings of the Annual Meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education* (2023)

Poster Presentations

Izge Bayyurt and Justin K. Dimmel. "Surface Explorer: Investigating 3D Solids Through Large-Scale Surfaces in Virtual Reality". In: *Proceedings of the Annual Meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education* (2024)

Teaching Experience

Graduate Student Lecturer

The University of Maine

Course: MAT 107 Elementary Descriptive Geometry

Spring 2024

Delivered lectures twice a week to education majors. Prepared slides, assignments, and exams.

Graduate Teaching Assistant

The University of Maine

Course: MAT 122 Precalculus

Fall 2023

Ran 3 recitations where students practiced concepts from the lectures using active learning methods and covered the topics that weren't touched during lectures. Graded exams and held office hours.

Teaching Assistant

Colby College

Courses: CS 151, 152, 231, and 375

Spring 2019 - Spring 2022

Held weekly lab hours to help other students with their project assignments and taught programming basics. Graded student work and worked with professors to design homeworks.

Work Experience

President, RiSE Research Group

January 2024 - Present

The University of Maine

- Scheduled meetings, planned presentations, coordinated with other students and reached out to faculty to have as guests speakers.

Junior Software Engineer

July - August 2022

Pocket Gems

- Worked as a software engineer for the backend team developing mobile video games.
- Used AWS tools to integrate 3rd party APIs to our system.

Software Engineering Intern

Copenhagen, Denmark

MeetinVR

October 2021 - February 2022

- Worked voluntarily as a part of a start-up focusing on developing virtual meeting rooms in VR.
- Enhanced existing software features, such as allowing sticky notes to have custom images to be used as background.
- Consulted with our partner companies who requested custom made meeting rooms.

Information Technologies Services Support

September 2018 - May 2022

- Worked as an IT support to assist students and faculty with installing various software and troubleshooting devices.

Personal Projects

ARTengri

Summer 2019

Game Designer, Quality Assurance

- Helped with the design of the AR card game created by Serkan Terlec. Using iterative design process, we made the game more fast-paced and entertaining.

HackUMass

October 12-14 2018

Programmer, Designer

- Developed an Android application which lets users see what people around them listen to through Spotify API. Assisted with the integration of the API, designed the UI and the app logo.

Bowdoin Hackathon

February 24 2019

Programmer, Designer

- Created an 2D arcade game in Unity with Tyler Hansen. Programmed the mechanics of the game and created sprites in Aseprite & Illustrator. Composed the main title and in-game tracks.
- Won 2nd place award for the best game

Other involvements

- Graduate student senator (2023 - present)
- UMGWU-UAW union organizer (2024 - present)
- Colby Music Incubator (2018-2021)